

BABU BANARASI DAS UNIVERSITY

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**SCHOOL OF COMPUTER
APPLICATION**

No SQL

**Project On: - Students Management
System Using MongoDB**

SUBMITTED BY: -

**Name: Arpita
Kushwaha**

Branch: BCA DS\AI

**University roll no:
1250258132**

SUBMITTED TO: -

Mr. Harendra Singh

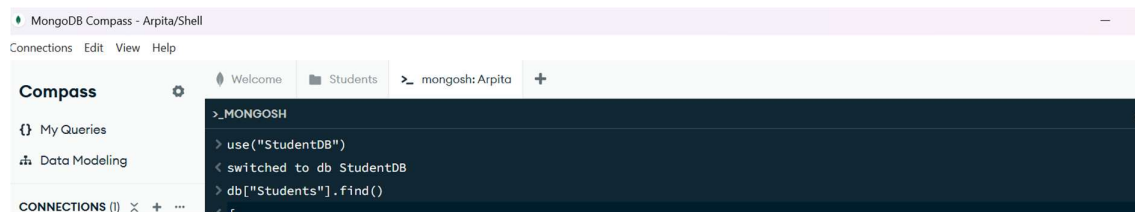
```
> db.students.countDocuments()  
< 45  
StudentDB>
```

1.CREATE DATABASE

MongoDB Query:

Use Students

Explanation: Creates or switches to the Student database.

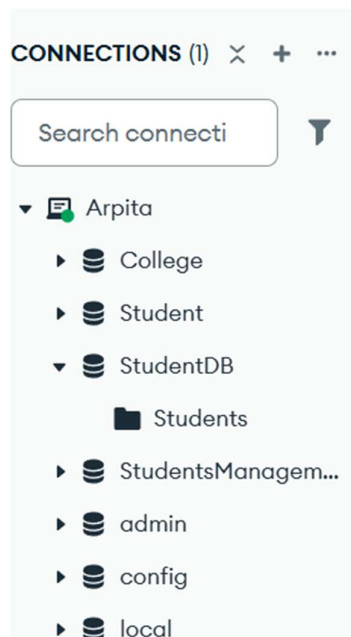


2.CREATE COLLECTION

MongoDB Query:

db. Create Collection("students")

Explanation: Creates a collection named Student.



3. INSERT 50 STUDENTS

MongoDB Query:

db.students.insertMany([[]])

Explanation: Insert multiple student documents into the collection.

```
{roll:1, name:"Riya", age:19, marks:84, city:"Lucknow"},
{roll:2, name:"Mohit", age:18, marks:72, city:"Kanpur"},
{roll:3, name:"Kavya", age:19, marks:91, city:"Varanasi"},
{roll:4, name:"Deepak", age:20, marks:67, city:"Gonda"},
{roll:5, name:"Pooja", age:19, marks:88, city:"Lucknow"},
{roll:6, name:"Siddharth", age:18, marks:79, city:"Prayagraj"},
{roll:7, name:"Ishita", age:20, marks:93, city:"Lucknow"},
{roll:8, name:"Harsh", age:19, marks:61, city:"Ayodhya"},
{roll:9, name:"Muskan", age:18, marks:86, city:"Kanpur"},
{roll:10, name:"Rajat", age:20, marks:74, city:"Jaunpur"},
{roll:11, name:"Tanvi", age:19, marks:95, city:"Lucknow"},
{roll:12, name:"Yash", age:18, marks:70, city:"Bareilly"},
{roll:13, name:"Sneha", age:19, marks:82, city:"Varanasi"},
{roll:14, name:"Ankit", age:20, marks:58, city:"Gorakhpur"},
{roll:15, name:"Palak", age:18, marks:89, city:"Prayagraj"},
{roll:16, name:"Mayank", age:19, marks:76, city:"Lucknow"},
{roll:17, name:"Divya", age:20, marks:90, city:"Kanpur"},
{roll:18, name:"Saurabh", age:18, marks:64, city:"Gonda"},
{roll:19, name:"Khushi", age:19, marks:87, city:"Lucknow"},
{roll:20, name:"Nitin", age:20, marks:69, city:"Ayodhya"},
{roll:21, name:"Ritika", age:18, marks:83, city:"Jaunpur"},
{roll:22, name:"Abhay", age:19, marks:77, city:"Kanpur"},
{roll:23, name:"Simran", age:20, marks:92, city:"Lucknow"},
{roll:24, name:"Gaurav", age:18, marks:55, city:"Bareilly"},
```

{roll:25, name:"Bhavna", age:19, marks:85, city:"Varanasi"},
{roll:26, name:"Piyush", age:20, marks:73, city:"Lucknow"},
{roll:27, name:"Aarti", age:18, marks:88, city:"Gorakhpur"},
{roll:28, name:"Manish", age:19, marks:66, city:"Prayagraj"},
{roll:29, name:"Komal", age:20, marks:94, city:"Lucknow"},
{roll:30, name:"Rishabh", age:18, marks:81, city:"Kanpur"},
{roll:31, name:"Payal", age:19, marks:75, city:"Ayodhya"},
{roll:32, name:"Karan", age:20, marks:62, city:"Lucknow"},
{roll:33, name:"Nandini", age:18, marks:90, city:"Jaunpur"},
{roll:34, name:"Shivam", age:19, marks:78, city:"Varanasi"},
{roll:35, name:"Garima", age:20, marks:96, city:"Prayagraj"},
{roll:36, name:"Tushar", age:18, marks:59, city:"Kanpur"},
{roll:37, name:"Isha", age:19, marks:84, city:"Lucknow"},
{roll:38, name:"Vikas", age:20, marks:71, city:"Gonda"},
{roll:39, name:"Chandni", age:18, marks:86, city:"Lucknow"},
{roll:40, name:"Arjun", age:19, marks:68, city:"Bareilly"},
{roll:41, name:"Rashmi", age:20, marks:91, city:"Kanpur"},
{roll:42, name:"Lokesh", age:18, marks:80, city:"Ayodhya"},
{roll:43, name:"Sakshi", age:19, marks:89, city:"Varanasi"},
{roll:44, name:"Devansh", age:20, marks:65, city:"Lucknow"},
{roll:45, name:"Mahima", age:18, marks:82, city:"Jaunpur"},
{roll:46, name:"Pankaj", age:19, marks:57, city:"Gorakhpur"},
{roll:47, name:"Anjali", age:20, marks:87, city:"Lucknow"},
{roll:48, name:"Raunak", age:18, marks:74, city:"Prayagraj"},
{roll:49, name:"Jyoti", age:19, marks:93, city:"Kanpur"},
{roll:50, name:"Sumit", age:20, marks:76, city:"Lucknow"}

```
db.students.insertMany([
  {roll:1, name:"Riya", age:19, marks:84, city:"Lucknow"},
  {roll:2, name:"Mohit", age:18, marks:72, city:"Kanpur"},
  {roll:3, name:"Kavya", age:19, marks:91, city:"Varanasi"},
  {roll:4, name:"Deepak", age:20, marks:67, city:"Gonda"},
  {roll:5, name:"Pooja", age:19, marks:88, city:"Lucknow"},
  {roll:6, name:"Siddharth", age:18, marks:79, city:"Prayagraj"},
  {roll:7, name:"Ishita", age:20, marks:93, city:"Lucknow"},
  {roll:8, name:"Harsh", age:19, marks:61, city:"Ayodhya"},
  {roll:9, name:"Muskan", age:18, marks:86, city:"Kanpur"},
  {roll:10, name:"Rajat", age:20, marks:74, city:"Jaunpur"},
  {roll:11, name:"Tanvi", age:19, marks:95, city:"Lucknow"},
  {roll:12, name:"Yash", age:18, marks:70, city:"Bareilly"},
  {roll:13, name:"Sneha", age:19, marks:82, city:"Varanasi"},
  {roll:14, name:"Ankit", age:20, marks:58, city:"Gorakhpur"},
  {roll:15, name:"Palak", age:18, marks:89, city:"Prayagraj"},
  {roll:16, name:"Mayank", age:19, marks:76, city:"Lucknow"},
  {roll:17, name:"Divya", age:20, marks:90, city:"Kanpur"},
  {roll:18, name:"Saurabh", age:18, marks:64, city:"Gonda"},
  {roll:19, name:"Khushi", age:19, marks:87, city:"Lucknow"},
  {roll:20, name:"Nitin", age:20, marks:69, city:"Ayodhya"},
  {roll:21, name:"Ritika", age:18, marks:83, city:"Jaunpur"},
])
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.students.find()
```

Explanation: Retrieves documents from the students collection.

```

db["Students"].find()
{
  _id: ObjectId('6ac116cfc7a7c75b5262c50b'),
  roll: 1,
  name: 'Riya',
  age: 19,
  marks: 84,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac116cfc7a7c75b5262c50c'),
  roll: 2,
  name: 'Mohit',
  age: 18,
  marks: 72,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ac116cfc7a7c75b5262c50d'),
  roll: 3,
  name: 'Kavya',
  age: 19,

```

5. EQUAL TO - \$eq

MongoDB Query:

db.students.find({marks: 88})

Explanation: Finds students whose marks are exactly 88.

```

> db.Students.find({marks:88})
< {
  _id: ObjectId('6ac116cfc7a7c75b5262c50f'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac116cfc7a7c75b5262c525'),
  roll: 27,
  name: 'Aarti',
  age: 18,
  marks: 88,
  city: 'Gorakhpur'
}

```

6. GREATER THAN - \$gt

MongoDB Query:

```
db.students.find({marks:{$gt:85}})
```

Explanation: Find students whose marks are greater than 85.

```
> db.students.find({marks:{$gt:85}})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd6'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 91,
  city: 'Varanasi'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bda'),
  roll: 7,
  name: 'Ishita',
  age: 20,
```

7. LESS THAN - \$lt

MongoDB Query:

```
db.students.find({marks:{$lt:60}})
```

Explanation: Finds students whose marks are less than 60.

```
> db.students.find({marks:{$lt:60}})
< {
  _id: ObjectId('6ac11e571aa23852c4247be1'),
  roll: 14,
  name: 'Ankit',
  age: 20,
  marks: 58,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6ac11e571aa23852c4247beb'),
  roll: 24,
  name: 'Gaurav',
  age: 18,
  marks: 55,
  city: 'Bareilly'
}
```

8. GREATER THAN OR EQUAL TO - \$gte

MongoDB Query:

```
db.students.find({marks:{$gte:90}})
```

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks:{$gte:90}})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd6'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 91,
  city: 'Varanasi'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bda'),
  roll: 7,
  name: 'Ishita',
  age: 20,
  marks: 93,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bde'),
  roll: 11,
  name: 'Tanvi',
  age: 19,
  marks: 95
}
```

9. LESS THAN OR EQUAL TO - \$lte

MongoDB Query:

```
db.students.find({marks:{$lte:70}})
```

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks:{$lte:70}})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd7'),
  roll: 4,
  name: 'Deepak',
  age: 20,
  marks: 67,
  city: 'Gonda'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bdb'),
  roll: 8,
  name: 'Harsh',
  age: 19,
  marks: 61,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bdf'),
  roll: 12,
  name: 'Yash',
  age: 18,
}
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne({roll:5},{set:{marks:95}})
```

Explanation: Updates the marks to 95 of the student with roll number 5.

```
> db.students.updateOne({roll:5},{set:{marks:95}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.find({roll:5})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.students.deleteOne({roll:10})
```

Explanation: Deletes one document matching roll number 10.

```
> db.students.deleteOne({roll:10})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.students.find({roll:10})
<
StudentDB >
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({marks:{<60}})
```

Explanation: deletes all students having marks below 60.

```
> db.students.deleteMany({marks:{<60}})
< {
  acknowledged: true,
  deletedCount: 4
}
> db.students.find({marks:{<60}})
<
StudentDB> |
```

13. AND OPERATION- \$and

MongoDB Query:

```
db.students.find({$and:[{marks:{>80}},{age:19}]})
```

Explanation: Finds students satisfying both conditions.

```
> db.students.find({$and:[{marks:{>80}},{age:19}]})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd4'),
  roll: 1,
  name: 'Riya',
  age: 19,
  marks: 84,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd6'),
  roll: 3,
  name: 'Kavya',
  age: 19,
  marks: 91,
  city: 'Varanasi'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
```

14: OR OPERATION - \$or

MongoDB Query:

```
db.students.find({$or:[{city:"Lucknow"},{city:"Kanpur"}]})
```

Explanation: Finds students from Lucknow or Kanpur.

```
> db.students.find({$or:[{city:"Lucknow"},{city:"Kanpur"}]})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd4'),
  roll: 1,
  name: 'Riya',
  age: 19,
  marks: 84,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd5'),
  roll: 2,
  name: 'Mohit',
  age: 18,
  marks: 72,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
```

15. SORT OPERATION

A. MongoDB Query:

```
db.students.find().sort({marks:1})
```

Explanation: Sorts students by marks from lowest to highest .use 1 for ascending order.

```
> db.students.find().sort({marks:1})
< {
  _id: ObjectId('6ac11e571aa23852c4247bdb'),
  roll: 8,
  name: 'Harsh',
  age: 19,
  marks: 61,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bf3'),
  roll: 32,
  name: 'Karan',
  age: 20,
  marks: 62,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247be5'),
  roll: 18,
  name: 'Saurabh',
  age: 18,
```

B. MongoDB Query:

```
db.students.find().sort({marks:-1})
```

Explanation: Sorts students by marks from lowest to highest .Use -1 for descending order.

```
> db.students.find().sort({marks:-1})
< {
  _id: ObjectId('6ac11e571aa23852c4247bf6'),
  roll: 35,
  name: 'Garima',
  age: 20,
  marks: 96,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bde'),
  roll: 11,
  name: 'Tanvi',
  age: 19,
```

16. LIMIT OPERATION

MongoDB Query:

```
db.students.find().limit(5)
```

Explanation: Display only the first five documents.

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6ac11e571aa23852c4247bd4'),
  roll: 1,
  name: 'Riya',
  age: 19,
  marks: 84,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd5'),
  roll: 2,
  name: 'Mohit',
  age: 18,
  marks: 72,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd6'),
  roll: 3,
  name: 'Kavya',
  age: 19,
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({city:"Lucknow"})
```

Explanation: Finds all students whose city is Lucknow.

```
> db.students.find({city:"Lucknow"})
< {
  _id: ObjectId('6ac11e571aa23852c4247bd4'),
  roll: 1,
  name: 'Riya',
  age: 19,
  marks: 84,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bd8'),
  roll: 5,
  name: 'Pooja',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ac11e571aa23852c4247bda'),
  roll: 7,
  name: 'Ishita',
  age: 20,
```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.students.countDocuments()
```

Explanation: Counts Documents currently present in the collection (performing after the delete function).

```
> db.students.countDocuments()
< 45
StudentDB >
```

